Streamlining Parameter Tuning in Full-Body Racing Simulators with an Automated Pipeline

Ruslan Shaiakhmetov[1(✉)], Danilo Pianini[1], Gabriele D'Angelo[1], and Valter Venusti[2]

[1] Alma Mater Studiorum—Università di Bologna, Cesena, Italy
{ruslan.shaiakhmetov,danilo.pianini,g.dangelo}@unibo.it
[2] Dallara Automobili S.p.A., Parma, Italy
v.venusti@dallara.it

Abstract. Simulation tools are essential for designing and testing car setups in modern racecar competitions: state-of-the-art full-body simulators replicate driving conditions, but they require detailed tuning (*alignment*) of hundreds or thousands of parameters to reduce the reality gap. This procedure is performed by comparing simulations with real-world data (Formula 1 teams collect data in the order of terabytes every race weekend), most often visually, resulting in a time-consuming operation that requires advanced expertise. Additionally, the process is not formally encoded, relies on human intuition and expertise, and thus results are highly subjective and may vary depending on who is performing the operation. In this paper, we present an automated pipeline for parameter tuning in full-body racing simulators. This pipeline replicates the manual tuning workflow, but substituting subjective visual comparisons with an objective cost function—the Residual Sum of Squares (RSS). We validate the proposed approach by comparing the quality of the alignment and the time required to achieve it between experts performing manual tuning and the proposed automated pipeline. We find that automating the process requires a similar number of simulations to be performed compared to manual tuning, but the automated pipeline is significantly faster and more consistent.

Keywords: Software automation · Data analysis · Optimization

1 Introduction

Besides attracting millions of fans globally, race car competitions have always been a hotbed of innovation. Technological advancements that derive from these

This work is supported by the project "Ottimizzazione computazione di un modello di simulazione dinamica del veicolo" funded by the European Union - NextGenerationEU and Dallara Automobili S.p.A. through the Italian "National Recovery and Resilience Plan" (PNRR) Mission 4, Component 2, Investment 3.3 (DM 352/2022) - CUP J33C22001400009.

Published by Springer Nature Switzerland AG 2025
H. Hojjat and G. Caltais (Eds.): FSEN 2025, LNCS 15593, pp. 155–169, 2025.
https://doi.org/10.1007/978-3-031-87054-5_11

activities led to improvements on commercial vehicles (drag reduction, active suspensions, turbochargers, dual-clutch transmissions, and so on), making car races become more than just a sport, and transforming them into laboratories that test cutting-edge technologies. With the advent of digitalization and data collection technologies, the motorsport industry has seen a significant shift in how racing cars are designed, optimized, and raced. Cars have thus progressively become more complex, featuring thousands of parameters that can be adjusted to optimize performance and efficiency. Additionally, cost-reduction actions [11] and environmental concerns [24] led to a progressive reduction in the number of physical tests allowed by the regulations.

Overall, these changes made simulation, and in particular sophisticated full-body simulators, a tool essential to be competitive: the car configuration can be tested and experimented with ahead of time in a virtual environment, allowing engineers to get to the track with a setup that is already close to optimal, saving precious time and resources. Doing so, however, requires simulations to be as close as possible to reality, in other words, it is paramount to reduce the *reality gap*. In reality, however, the on-track behavior of a race car often diverges from predictions based on CAD models. In fact, many interdependent factors can influence car performance dramatically: environmental conditions (temperature, tarmac roughness, kerbs shape, wind, ...), mechanical and aerodynamic properties (tire lateral and longitudinal grip, body stiffness, aerodynamic drag, ...), and car setup (gear ratios, engine map, brake balance, flap incidence, ...). Although some of the parameters can be measured directly with precision, others can at most be educated guesses which most often fall short of reality: among the others, tire stiffness, aerodynamic drag, and actual engine power are frequently misaligned with their real-world values. This reality gap needs to be filled if simulation is to be used to find the optimal setup. This tuning process is an optimization problem, aimed at finding the optimal set of parameters that minimize the discrepancy between simulated and real-world car behavior.

The first step in this optimization is the collection of extensive real-world data. After a race weekend, Formula 1 engineers typically gather around 1.5 TB of data, collected from modern race cars equipped with up to 300 sensors.[1] Filtered, cleaned, and processed data can then be used to configure a full-body racing simulator by replicating real-world conditions, including the track, weather, and car setup. Typically, then, a professional driver drives the simulator, by sitting in a cockpit that replicates the real car's interior, generating real-time data of the simulated car's behavior. A picture of one such simulators is shown in Fig. 1. Afterward, engineers compare simulation results with actual race data, identifying discrepancies to adjust the parameters accordingly. This iterative process of identifying gaps and adjusting the simulator's parameters continues until the simulation mimics the real-world performance as closely as possible.

Currently, the tuning process in full-body racing simulators remains predominantly manual, requiring significant time and effort from engineers. This method consumes considerable resources, it is heavily reliant on the expertise and intu-

[1] https://archive.is/lEmIf.

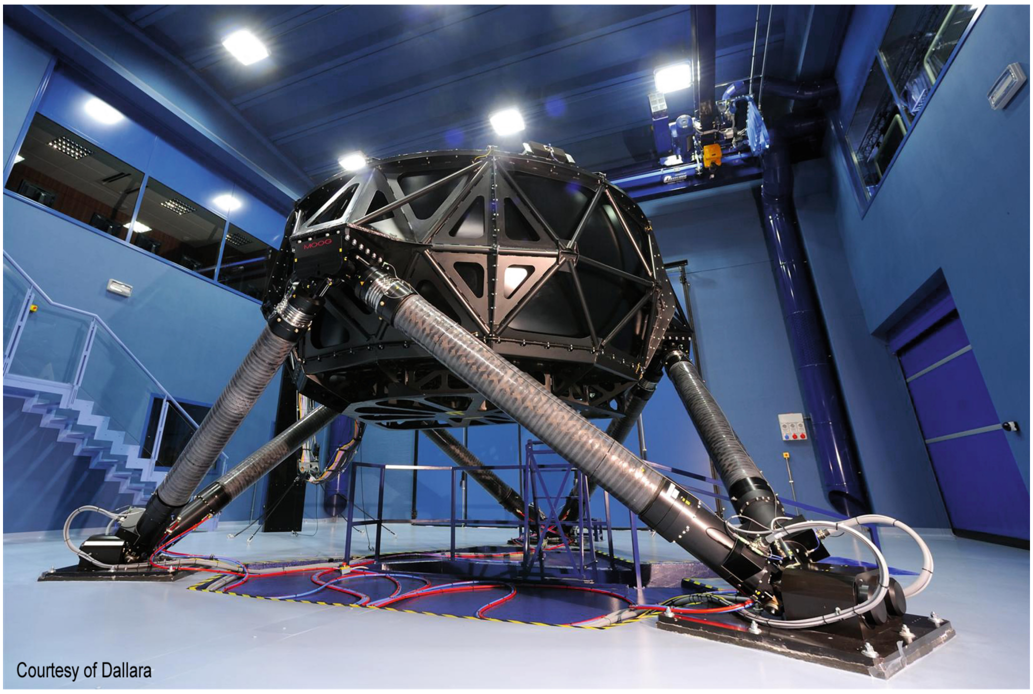

Fig. 1. Full motion racing simulator of Dallara Automobili S.p.A.

ition of engineers, and lacks consistency, making it difficult to compare results across different simulators or test environments. Moreover, while this specialized knowledge can sometimes lead to quicker solutions, it also carries the risk of misinterpretation, steering the optimization process in the wrong direction.

To complicate the problem at hand further, strong interdependencies between parameters may lead to quasi-optimal solutions, with multiple parameter configurations producing similar performance outcomes in peak performance, but with different resilience to changing driving styles and environmental conditions.

Ultimately, the combination of manual alignment, human factors, and complex parameter relationships can lead to suboptimal results. Different engineers working independently on the same simulator are likely to produce diverse configurations, which in turn produces misaligned performance and feedback across simulations. This lack of reproducibility hinders benchmarking and cross-comparisons, further complicating the development and refinement of simulators and car upgrades.

Contribution. In this work, we propose an automated pipeline to streamline the parameter alignment process in full-body racing simulators, improving efficiency, consistency, precision, and reproducibility.

Structure of the Paper. The remainder of this work is organized as follows: Sect. 2 provides an overview of existing approaches for parameter alignment in racing simulators, including software tools and methodologies; Sect. 3 introduces the automated pipeline designed to streamline the alignment process; Sect. 4 presents the experimental evaluation of the pipeline, comparing its performance with manual alignment; finally, Sect. 5 summarizes the results and discusses future directions.

2 State of the Art

To the best of our knowledge, no existing work specifically addresses the automation of the parameter alignment process between full-body racing simulators and real-world data. Unfortunately, the motorsport industry favors trade secrets [23] over patents and open-source solutions, keeping proprietary knowledge in-house. As such, some methods that are currently part of the state of the practice may not be publicly available. Thus, since the current state of the art in parameter alignment for racing simulators largely depends on manual methods, the following sections will examine these approaches. We will also review existing software tools designed for manual alignment analysis, and explore solutions that leverage Artificial Intelligence (AI) and Machine Learning (ML) for data analysis.

2.1 Proprietary Software Tools

Manual alignment approaches are so widespread that a range of specialized software solutions for race data acquisition and analysis have been developed to support them. Among the most popular tools for professional use are MoTec i2[2] and ATLAS[3], both widely utilized in motorsports for advanced telemetry analysis. These tools are complemented by more accessible options such as Cosworth PI Toolbox, AIM Race Studio, and VBOX Motorsport, which cater to a broader audience, including amateur racing teams and enthusiasts. These software tools are designed to quickly analyze vast amounts of data collected from race cars and offer comprehensive toolchains for in-depth data analysis.

Often, these tools are integrated into hardware-software ecosystems. For example, ATLAS is part of McLaren Applied Technologies, originally developed for Formula 1 cars. It is deeply integrated with the car's telemetry system, enabling real-time data acquisition and analysis, and connects to a cluster-based data storage infrastructure. This close integration ensures that the telemetry data is processed efficiently, allowing engineers to make rapid adjustments during races or testing sessions.

2.2 Artificial Intelligence-Based Approaches

There exists a distinct category of approaches designed to simulate the behavior of real cars in competitive racing games. These solutions leverage machine learning algorithms [7] or artificial intelligence techniques [9] to learn and replicate

[2] https://archive.is/tAoTG.
[3] https://archive.is/7OZsK.

the behavior of real cars and their drivers, using real-world data. Despite the high accuracy of the results [8], these solutions fail to provide explicit information about the car's parameters, limiting their usefulness when trying to evolve the car's configuration.

The rise of artificial intelligence and machine learning has significantly impacted the motorsport industry [16], primarily in real-time data analysis and anomaly prediction [26]. AI can learn a car's nominal behavior and predict anomalies in real-time, signaling the need for pit stops.

In offline applications, AI is utilized for team management [2], but it has yet to be fully integrated into race car design and engineering. However, as Matt Harris (Head of IT in Mercedes AMG at the time the declaration was registered) points out, finding the right balance between data analysis and human intuition is key to achieving competitive success [16]. Thus, manual data processing remains the primary tool for data analysis in motorsport [13], and data analysts sift through vast amounts of data to uncover insights that are not immediately apparent.[4]

2.3 Data Analysis: Metrics and Tools

Direct comparison of real-world vehicle data with simulated data often does not produce meaningful results due to the high level of uncertainty and interdependence of the variables involved in the problem. To address this challenge, engi-

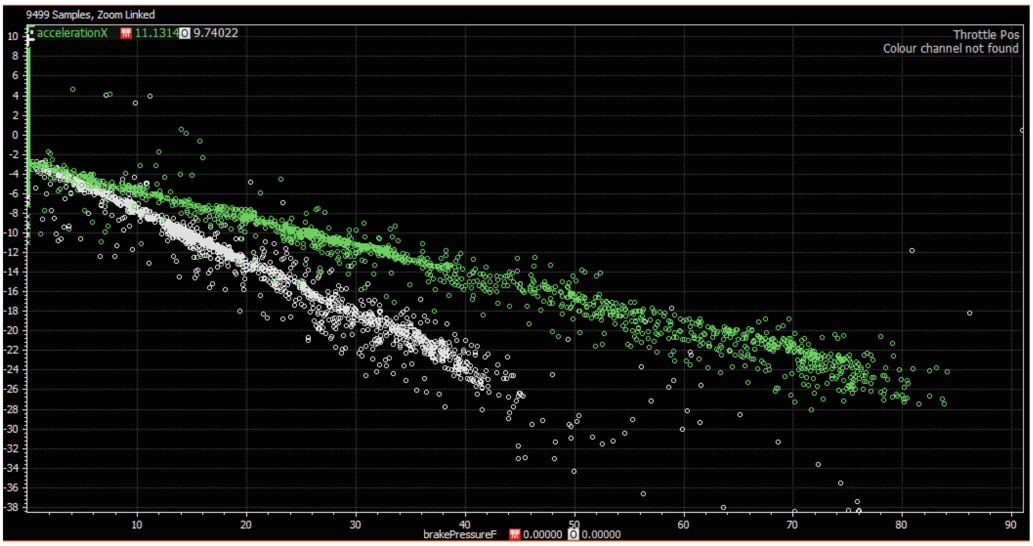

Fig. 2. Comparison of real-world and simulated data for brake efficiency (acceleration vs. brake pressure when acceleration is negative) in the interface of Motec i2 Pro. White dots represent real-world data, while green dots represent simulated data. In this case, the simulation underestimates the brake efficiency (lower deceleration for the same brake pressure). The closer real-world data and simulated data are, the smaller is the abstraction gap, and the more accurate is the alignment. (Color figure online)

[4] https://archive.is/GD4JK.

neers have introduced many *metrics* [4,6,12,19,21,27]. In simple terms, *metrics* are guidelines or methods designed to identify data mismatches that correspond to specific parameter discrepancies. This approach reduces uncertainty and helps pinpoint which parameters need adjustments and in which direction.

Typically, although data counts thousands of different telemetry data sources (in short, *channels*), only a few are selected for comparison, most often two, because they are easier to visualize and analyze manually than higher-dimensional data. Indeed, the previously mentioned metrics are, in most cases, designed to compare two channels at a time. These data is then often filtered to exclude irrelevant information, and then used to deduce properties of the car. A typical example is the comparison of velocity and acceleration (channels) when the throttle is fully open (filter), which provides information about the car's aerodynamic drag. Another example is the evaluation of the breaking efficiency, in which acceleration and pressure on the brake pedal are compared, filtering only data where acceleration is negative. Once the channel data is ready, it gets plotted for both the real-world and simulated data, and compared visually to identify discrepancies. Figure 2 shows a sample of brake efficiency evaluation from Motec i2 Pro.

In some cases, a mathematical (virtual) channel, automatically generated based on data from other channels using a mathematical equation, is used for data comparison. A common example is the understeer[5] gradient [5], which is derived from multiple variables, including tyre cornering stiffness and lateral forces.

2.4 Limitations of Manual Alignment

Limitations of manual alignment typically stem from human factors, and can be categorized into *recurrent* and *occasional*. Recurrent issues include the following.

- **Interdependent parameters.** Human engineers can plot and observe (hence, investigate) a limited number of parameters at once. However, oftentimes interdependencies across multiple parameters lead to iterative procedures, as changes in a small subset of parameters may affect others, requiring further adjustments and prolonging the alignment process.
- **Over-reliance on familiar solutions**. Experienced engineers may rely too heavily on previous experience, and thus, especially with novel car models, may overlook new, better solutions coming from unpredicted interactions. A notable example has been porposing in 2022 ground-effect based Formula 1 cars [28], largely unpredicted before taking the cars on track.
- **Round numbers as goals**. Humans like to deal with round numbers [17], which are simpler to remember and to work with, but this behavior introduces biases [10,25].
- **Greedy search**. Manual alignment tends to be more greedy than automatic algorithms, which typically include both exploration and exploitation phases.

[5] When understeering, a turning car rotates less than the driver commands.

Performing exploration, however, requires additional computational steps, which is time consuming and hardly feasible in manual alignment.

Occasional problems concerns most often human factors, such as tiredness, and include:

- **Lost memory of previous simulations**. Humans may forget results from previous simulations, leading to unnecessarily repeated work, extended alignment time, and increased computational costs.
- **Configuration mistakes**. Tools that support manual alignment are often complex, and it is relatively simple to make mistakes in configuration entries. Typographical errors, misused punctuation, and similar issues can lead to incorrect setups and wasted computation, prolonging the alignment process. Additionally, this kind of error is subtle, as they are difficult to spot and may lead to apparently reasonable results, thus making debugging hard.

3 An Automated Pipeline for Parameter Alignment

To streamline the parameter alignment process in full-body racing simulators, we propose an automated pipeline (depicted in Fig. 3) that transforms the traditionally manual alignment workflow into a fully automated system.

The pipeline requires *reference data* to align a virtual car model with. Although designed to work with real-world data, the pipeline can also accommodate virtual data sources: indeed, the goal is to find the set of parameters that minimizes the discrepancy between the simulated model and the target data, regardless of how the latter was obtained. A key challenge to do so is quantifying such discrepancy, which in manual alignment is typically done visually. In other words, we need a well-defined *cost function*. The cost function can be arbitrarily complex, and different cost functions can be devised and selected by configuring the pipeline. In this paper, we used the Residual Sum of Squares (RSS) between target and simulated data as cost function.

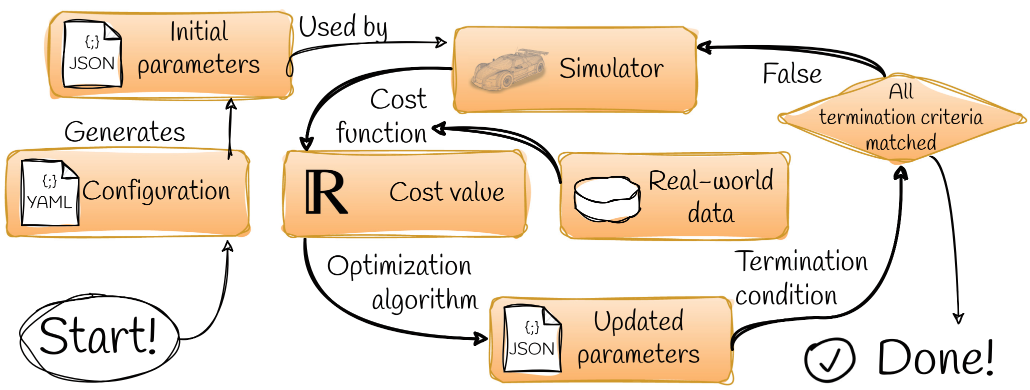

Fig. 3. Graphical depiction of the automated pipeline flowchart.

One critical aspect of comparing real-world and simulated channels (time series) is handling sample misalignment in time. For instance, one car sensor could provide readings at 30 Hz, while simulated data produces the same reading at 100 Hz.[6] In this case, real-world data needs to be resampled to match simulation data. Since some points in time may not exist in the real-world data, *interpolation* is required to fill in the gaps. By default, the pipeline uses linear interpolation, but this behavior is configurable.

3.1 Implementation Details

The software pipeline has been designed according to execute based on configuration. The configuration includes the elements depicted in Fig. 4, of which we will discuss the most relevant aspects.

Optimization is performed on `OptimizationUnits` ordered by priority (for same-priority units, the order is arbitrary). Each `OptimizationUnit` contains a set of interdependent `Parameters` whose optimization must be performed as a single multi-dimensional problem. In fact, high-end racecar models includes thousands of parameters, whose optimization is hardly treatable in a single shot due to the curse of dimensionality [1]. Consequently, parameters are grouped into `OptimizationUnits` based on their interdependencies (parameters very likely to affect each other are grouped together), and each unit is optimized separately from the others. Although some interdependencies among parameters are well-known, the general problem of creating groups of interdependent parameters which are independent of others requires an in-depth understanding of the car's physics, and some interdependencies can be discovered only by repeated experimentation.

`Parameters` have a validity range (for instance, negative drags make no physical sense), and a tolerance that can be used as termination condition for the optimization process. The quality of the current alignment for a specific parameter is determined by the weighted average of the cost function produced by

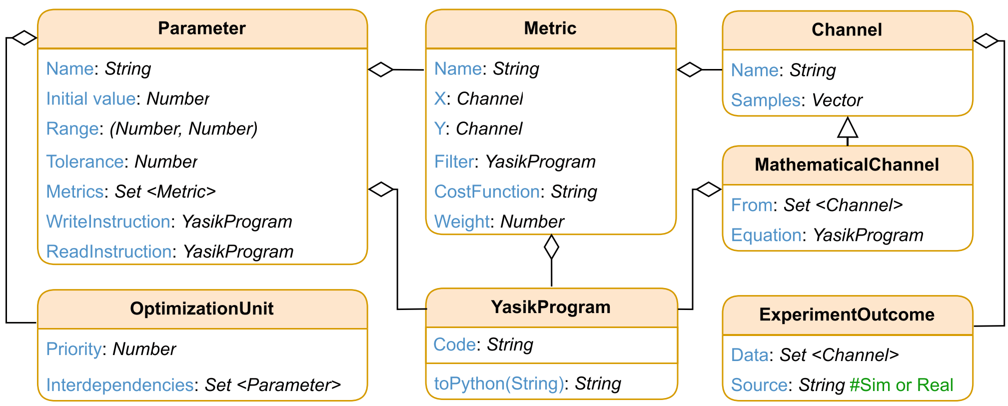

Fig. 4. UML class diagram showing the structure of the pipeline configuration entities.

[6] 100 Hz is the typical update frequency of state-of-the-practice full-body simulators.

each `Metric` associated to the parameter. At each optimization round, parameters are read from the simulator configuration, optimized, and then exported as configuration parameters of the simulator for the evaluation of their quality. Doing so often requires data manipulation to adapt the format of the parameters to the simulator's requirements, which is performed by scripts written in a custom Domain Specific Language (DSL) called Yasik specifically developed to express mathematical manipulations. Since most engineers in the motorsport industry are familiar with MATLAB, Yasik is a MATLAB-like DSL that compiles to Python. Yasik is released as open-source with a permissive license[7] and archived for future reference on Zenodo [18].

`Metrics` relate two `Channels` together, define a filter to select only the relevant data, and produce a cost function that measures how well the relationship between channels is reproduced by the simulator. Multiple channels can be combined into a `MathematicalChannel`. Data for channels comes from `ExperimentOutcomes`, which can be real-world or simulated data.

The pipeline has been implemented in Python. The software implementing the pipeline is a strategic industrial asset; as such, it is closed source and proprietary, and cannot be released as a companion artefact for this paper. Configuration is encoded in YAML[8], a human-friendly data serialization format.

Optimal parameter values are determined by minimizing the cost function, where each evaluation requires a simulation run, placing this problem within the Simulation-based optimization problem (SBOP) class. Selecting the best optimization algorithm is essential for achieving rapid convergence. In this work, we automate the entire alignment process using Nelder-Mead [14], also known as downhill simplex method, as it is simple yet quickly converging [3,15,20]. More complex scenarios may benefit from approaches with a different trade-off between exploration and exploitation, such as Bayesian optimization [22].

4 Experimental Evaluation

To evaluate the automated pipeline's performance, we compare it with manual alignment performed by three human engineers. We challenge both the automated pipeline and the human experts into performing the alignment process 10 times on the same real-world data, taken from a real-world racing event that took place in the Marina Bay Street Circuit (Singapore, September 2024). Data has been filtered with a low-pass filter to remove noise, resampled to 100 Hz and internally synchronized (time aligned). As car model, we adopt simpler model to speed up the simulation process, based on the well-known "single-track model", in which the car is approximated as a two-wheeled vehicle with a fixed wheelbase. The model in simulation is controlled by a state-of-the-art virtual driver following the same trajectory of the real-world driver.

[7] https://github.com/Crylab/yasik.
[8] https://yaml.org/spec/1.2.2/.

We try to align two parameters: the aerodynamic drag coefficient C_D and the end-to-end ratio of the 8th gear z_8; for each, we select two relevant channels. In contast to the plain gear ratio[9], which is well-known in advance (and, in some categories, fixed in the rulebook), the end-to-end gear ratio is a simulation parameter which measures the relationship between the engine's rotational speed and the car's speed; thus, it accounts for many real-world interactions (mechanical, such as friction; thermal, such as tyre temperature; electrical, such as engine maps; and contextual, such as tarmac shape and composition) across all components that are involved in the conversion of engine rotations to the car's speed. Although the aerodynamic drag coefficient and the end-to-end ratio of the 8$^{\text{th}}$ gear are a very small subset of the parameters that need to be aligned in a real-world scenario, they are particularly representative: the aerodynamic drag coefficient is typically the *first* parameter to be optimized (practical relevance), and the end-to-end gear ratio is inherently complex as it results from interdependencies of multiple parameters (demonstrating that the approach can deal with complexity). The experiment configuration is summarized in Table 1. In the first experiment, we ask engineers and configure the pipeline to align the aerodynamic drag coefficient C_D. In the second experiment, we ask them to align both C_D and the end-to-end 8$^{\text{th}}$ gear ratio z_8. We ask human engineers to perform *at most* 26 and 50 iterations for the first and second experiment, respectively. We do so to simulate the time constraints of a real-world scenario.

Table 1. Optimization Parameters and Metrics

Aspect	Aerodynamic Drag Coefficient (C_D)	End-to-end 8$^{\text{th}}$ gear ratio (z_8)
Initial Value	1.0	10.0
Range	$0.5 \ldots 3.0$	$5.0 \ldots 20.0$
Weight	$6.36 \cdot 10^{-3}$	$2 \cdot 10^{-9}$
Reference channels	Acceleration vs Speed	RPM vs Speed
Filter	$V_x > 67.0$ *and* $Th > 90\%$	Gear $>$ 7.5 and Gear $<$ 8.5 and Brake $<$ 0.5
Cost Function	LR[a] + RSS	LR + RSS

[a]Linear regression

[9] Formally, the rotational speed of the output shaft divided by the angular speed of the input shaft.

Since human beings can memorize the results of previous alignments, we had to devise a way to minimize inter-experimental bias. We do so by generating, for each manual alignment attempt, a hidden random shift for each provided parameter's value; for instance, if the drag coefficient C_D is among the parameters to get aligned, a random shift ρ_{C_D} is generated, and the value entered by the engineer α_{C_D} will actually produce, internally, the value $\alpha_{C_D} + \rho_{C_D}$. Equivalently, for the automated pipeline, we set randomized initial values for all parameters to be aligned within their validity ranges.

We use two metrics to evaluate the alignment process: 1. the **variance** of the final parameter values: the lower the variance, the more consistent the alignment process; and the **total time** required to reach alignment: the lower the time, the faster the alignment process.

4.1 Results

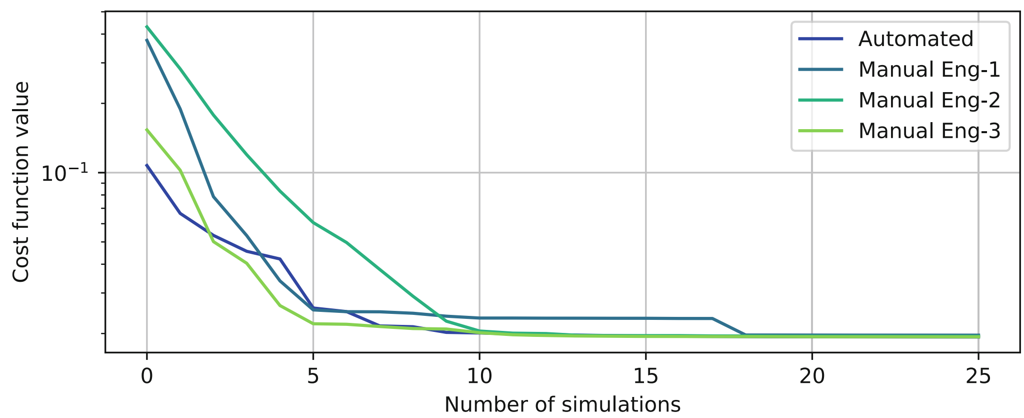

Fig. 5. Alignment quality of the aerodynamic drag coefficient C_D, averaged over 10 experiments, the lower the better. The automated tool (blue) shows a similar exponential decay in the metric value, replicating the manual approach automatically. (Color figure online)

The mean result quality for both human and automated alignment is depicted in Fig. 5. The automated pipeline demonstrates a convergence rate per iteration comparable to manual alignment, effectively replicating the manual approach. However, as shown in Fig. 6, the automated pipeline completes the optimization process nearly three times faster than manual alignment. The most time-consuming part of the process, apart from simulation, involves iterating over datasets to compute the RSS.

Additionally, the variance in the final parameter values is more then 2.5 times lower in the automated pipeline compared to manual alignment. While individual manual alignment sessions may sometimes yield more accurate results, the automated approach provides more consistent outcomes across different sessions.

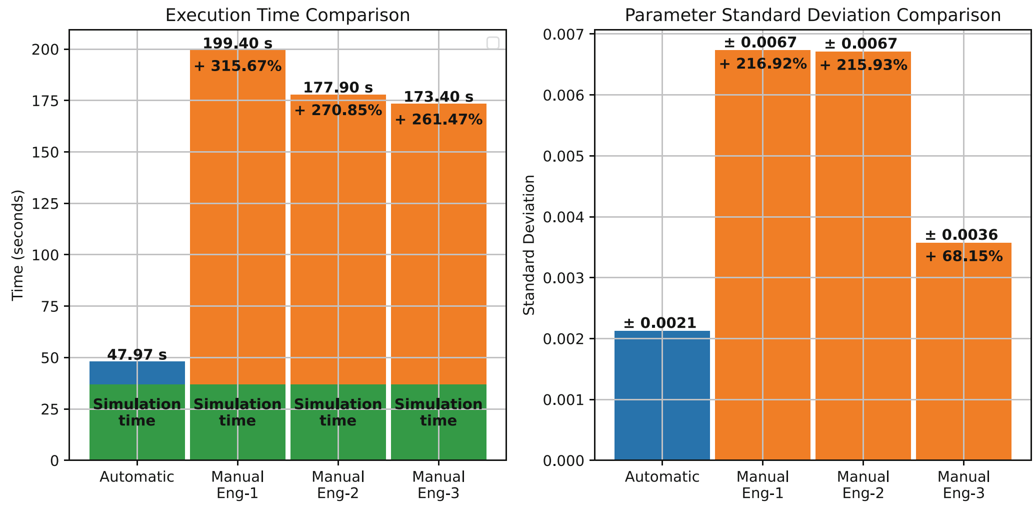

Fig. 6. Parameter variance and total alignment time of automated and manual alignment.

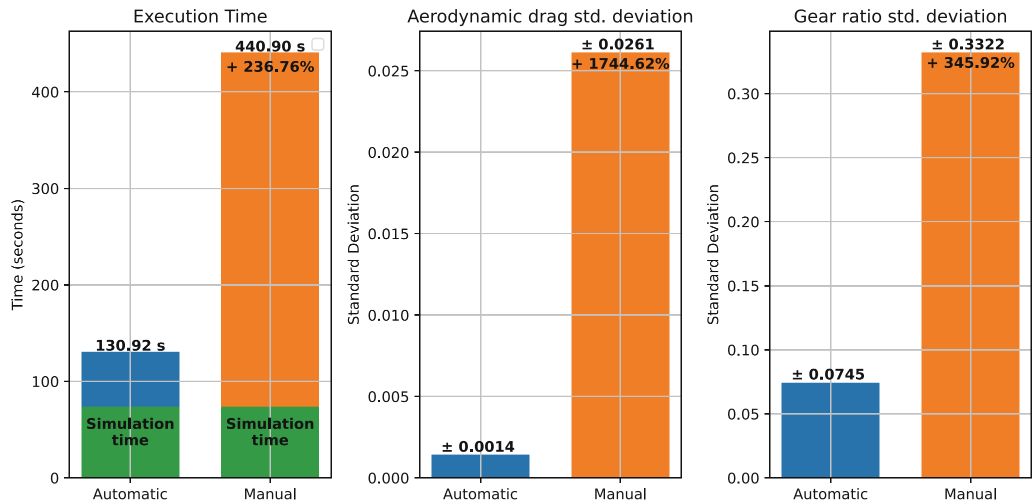

Fig. 7. Optimization of interdependent parameters.

Moreover, the automated pipeline is more effective at handling interdependent parameters than manual alignment. This is evident in the second experiment, where engineers adjusted the parameters iteratively, while the automated pipeline used a joint cost function and optimized both parameters simultaneously. The results of this experiment are shown in Fig. 7: within 50 iterations, the automated pipeline reduced variance of C_D to levels comparable to those of the previous experiment, while optimizing z_8 at the same time. When aligning manually, however, the extra parameter led to a four-folded increase in variance for both C_D and z_8.

Overall, the optimization time was twice shorter with the automated pipeline, demonstrating its ability to efficiently manage interdependent parameters while providing a more accurate alignment process in a shorter time frame.

5 Conclusion and Future Work

In this work, we presented an automated pipeline for parameter alignment in full-body racing simulators. Our approach stems from an analysis of the existing best practices to understand open challenges, sources of variability, and opportunities to improve. We discover that most of the current limitations arise from human factors and from the complexity of the problem at hand, in particular interactions among parameters. We then presented an automated pipeline meant to speed up the alignment process, discussing the key components and the design choices.

We evaluated the effectiveness of our solution by comparing the consistency (variance) and the time required to perform the alignment process between the automated pipeline and three human engineers. Data shows that the automated pipeline significantly outperformed manual alignment in both consistency and speed. Specifically, the automated process reduced parameter variance by more than 2.5 times, demonstrating far greater consistency and repeatability of the process. Additionally, the automated pipeline completed the optimization process in roughly one-third of the time required for manual alignment.

The benefits of automation were particularly evident when dealing with interdependent variables. In these scenarios, the automated solution preserved convergence properties despite the increased complexity and dimensionality of the problem, while trying to perform the operation manually registered a considerable decrease in quality: manual attempts required more than twice as much time, and often exacerbated variance.

Future work will focus on enhancing the closed-loop system for data collection, analysis, and automated adjustments, creating a more adaptive and responsive alignment framework. Additionally, we aim to explore the integration of a digital twin solution for race cars, enabling real-time analysis and alignment while real cars run on track. Such a system could continuously monitor vehicle performance during races, dynamically adjusting parameters to maximize performance and potentially providing a competitive edge in racing competitions.

Another future direction is to explore dividing the optimization process into "weakly interdependent" parameter groups. This approach could potentially accelerate the optimization of groups with high dimensionality (10 and above) by splitting them into smaller units, optimizing each separately, and exchanging parameters values between them at run-time. To be applied, however, this approach would require (i) advances in the optimization algorithms to account for different degrees of interdependency among subgroups and shared information; and (ii) a robust measure of interdependency, to be used to identify boundaries for splitting the problem and to balance the amount of information sharing.

Finally, both quality and execution time of the alignment process are tightly bound to the quality and representativeness of the simulated data. Thus, the role of the driver is crucial, yet professional drivers are scarce, and their time and feedback much more valuable *after* alignment has been performed to test and provide feedback with different car configurations. Consequently, we expect that further major steps in the automation of the alignment process will involve the development of better virtual driver.

References

1. Altman, N., Krzywinski, M.: The curse(s) of dimensionality. Nat. Methods **15**(6), 399–400 (2018)
2. Bouchet, A., Doellman, T.W., Troilo, M., Walkup, B.R.: Pre-empting the competition: how do shareholders view sponsorships in the sport apparel industry? J. Sport Manag. **31**(3), 275–287 (2017)
3. Bujari, D.: Hyperparameter optimization in recommender systems: Bayesian optimization vs. nelder-mead (2020). https://www.politesi.polimi.it/handle/10589/175652
4. Christopher, B.: Making Sense of Squiggly Lines: The Basic Analysis of Race Car Data Acquisition. CB-Racing (2011)
5. Ganzarolli, F., de C. Leal, A.T.J.: Influence of understeer gradient variation during cornering in the vehicle stability perception. In: SAE Technical Paper Series. BRASILCONG, SAE International (2014). https://doi.org/10.4271/2014-36-0209
6. Guiggiani, M.: The Science of Vehicle Dynamics: Handling, Braking, and Ride of Road and Race Cars. Springer (2014). https://doi.org/10.1007/978-94-017-8533-4
7. Hojaji, F., Toth, A.J., Campbell, M.J.: A machine learning approach for modeling and analyzing of driver performance in simulated racing. In: Longo, L., O'Reilly, R. (eds.) Artificial Intelligence and Cognitive Science - 30th Irish Conference, AICS 2022, Munster, Ireland, 8–9 December 2022, Revised Selected Papers. Communications in Computer and Information Science, vol. 1662, pp. 95–105. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-26438-2_8
8. Hojaji, F., Toth, A.J., Joyce, J.M., Campbell, M.J.: AI-enabled prediction of sim racing performance using telemetry data. Comput. Hum. Behav. Rep. **14**, 100414 (2024). https://doi.org/10.1016/j.chbr.2024.100414
9. Hojaji, F., Toth, A.J., Joyce, J.M., Campbell, M.J.: An AI approach for analyzing driving behaviour in simulated racing using telemetry data. In: Dondio, P., Rocha, M., Brennan, A., Schönbohm, A., de Rosa, F., Koskinen, A., Bellotti, F. (eds.) Games and Learning Alliance - 12th International Conference, GALA 2023, Dublin, Ireland, November 29 - December 1, 2023, Proceedings. Lecture Notes in Computer Science, vol. 14475, pp. 194–203. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-49065-1_19
10. Jain, G., Gaeth, G.J., Nayakankuppam, D., Levin, I.P.: Revisiting attribute framing: the impact of number roundedness on framing. Organ. Behav. Hum. Decis. Process. **161**, 109–119 (2020)
11. Kallimogiannis, V., Massouros, C.: Predictions on the impact of the budget cap on formula 1 teams' performance. Ratio Math. **47** (2023). https://doi.org/10.23755/rm.v47i0.1179
12. Knox, B.: A Practical Guide to Race Car Data Analysis. CreateSpace Independent Publishing Platform (2011)

13. Lopez, G., Seaber, A.: The theory and practice of race vehicle data acquisition and analysis in motor sports engineering education. In: 2009 Annual Conference I& Exposition Proceedings, pp. 14.1257.1–14.1257.17. ASEE Conferences (2009). https://doi.org/10.18260/1-2--4814
14. Nelder, J.A., Mead, R.: A simplex method for function minimization. Comput. J. **7**(4), 308–313 (1965)
15. Ozaki, Y., Yano, M., Onishi, M.: Effective hyperparameter optimization using nelder-mead method in deep learning. IPSJ Trans. Comput. Vis. Appl. **9**, 20 (2017). https://doi.org/10.1186/S41074-017-0030-7
16. Patel, D., Shah, D., Shah, M.: The intertwine of brain and body: a quantitative analysis on how big data influences the system of sports. Ann. Data Sci. **7**(1), 1–16 (2020)
17. Pope, D., Simonsohn, U.: Round numbers as goals: evidence from baseball, sat takers, and the lab. Psychol. Sci. **22**(1), 71–79 (2010)
18. Ruslan Shaiakhmetov, D.P.: Crylab/yasik: Release v. 0.0.7 by 2024-10-23-15:03:02 (2024). https://doi.org/10.5281/ZENODO.13983366. https://zenodo.org/doi/10.5281/zenodo.13983366
19. Segers, J.: Analysis Techniques for Racecar Data Acquisition, 2nd edn. SAE International, Warrendale (2014)
20. Senthil Vel, A., Konan, K.E., Cortes-Borda, D., Felpin, F.X.: Enhancing optimization of mixed variables on a robotic flow platform: integrating statistical filtering with nelder-mead and bayesian methods. Organic Process Res. I& Dev. **28**(5), 1597–1606 (2023)
21. Seward, D.: Race Car Design. Red Globe Press, London (2014)
22. Shahriari, B., Swersky, K., Wang, Z., Adams, R.P., de Freitas, N.: Taking the human out of the loop: a review of bayesian optimization. Proc. IEEE **104**(1), 148–175 (2016)
23. Solitander, M., Solitander, N.: The sharing, protection and thievery of intellectual assets: the case of the formula 1 industry. Manag. Decis. **48**(1), 37–57 (2010)
24. Sturm, D., Andrews, D.L., Miller, T., Bustad, J.: Green light or black flag? Greenwashing environmental sustainability in formula one and formula e. Ann. Leisure Res. 1–20 (2024). https://doi.org/10.1080/11745398.2024.2376754
25. Thomas, M., Park, J.: The precision effect: how numerical precision influences everyday judgments. SSRN Electron. J. (2013)
26. Tulabandhula, T., Rudin, C.: Tire changes, fresh air, and yellow flags: challenges in predictive analytics for professional racing. Big Data **2**(2), 97–112 (2014)
27. William F. Milliken, D.L.M.: Race Car Vehicle Dynamics. SAE International (1994)
28. Zhu, Z.: Cause and analysis of 2022 formula 1 'porpoising'. High. Sci. Eng. Technol. **46**, 19–27 (2023). https://doi.org/10.54097/hset.v46i.7659